PHYSICAL CHEMISTRY CUMULATIVE EXAM

November 16, 1996

[30] The isenthalpic Joule-Thomson coefficient for $CO_2(g)$ is μ_{JT} = 1.11 K bar^{-1} at 320K. Assuming this to be constant, predict the temperature change when CO_2 gas at 150 bar expands through an orifice to a final pressure of 1.00 bar, at constant enthalpy.

[60] The Joule-Thomson coefficient actually depends on both temperature and pressure. For a given starting temperature, the pressure dependence may be described by an equation of the form $\mu(p) = a + bp + cp^2$, where a, b, and c are empirical coefficients. Consider CO_2 gas initially at 320 K and 150 bar which expands through an opening to a final pressure of 1.00 bar. Predict the temperature change given that, for CO_2, the empirical coefficients are: a = 0.8187 K bar^{-1}, b = -3.276×10^{-3} K bar^{-2}, and c = -5.66×10^{-6} K bar^{-3}.

[10] CO_2 sublimes at -78°C under an applied pressure of 1 bar. During the expansions considered above, should the CO_2 solidify?